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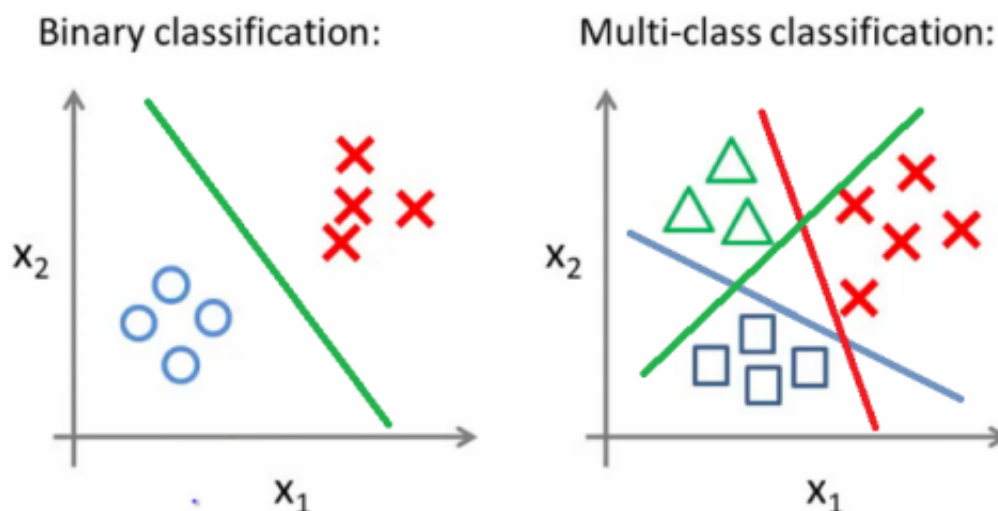
Examples of Logistic Regression

What is Logistic Regression?

- Logistic Regression is one of the **supervised machine learning algorithms** that need a good background knowledge of **calculus, probability & statistics**.
- The idea of Logistic Regression is to find a **relationship between features and probability of particular outcome**.
- Logistic regression is used when the response **you want to predict/measure** is **categorical** with **two or more levels**.

Types of Logistic regression

- **Binomial Logistic Regression**, where the response variable has two values 0 and 1 or pass and fail or true and false.



- **Multinomial Logistic Regression** deals with situations where the response variable can have three or more possible values.

Why Logistic, not Linear?

- With **binary classification**, let ' $\mathbf{x}$ ' be some feature and ' $\mathbf{y}$ ' be the output which can be either 0 or 1.
- The probability that the output is 1 given its input can be represented as:

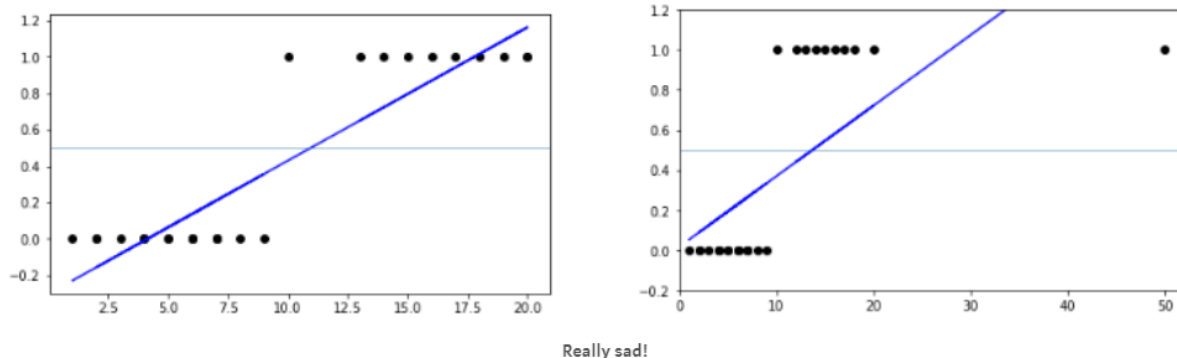
$$P(y = 1 | x)$$

- If we predict the **probability via linear regression**, we can state it as:

$$p(X) = \beta_0 + \beta_1 X.$$

where, $p(x) = p(y=1|x)$

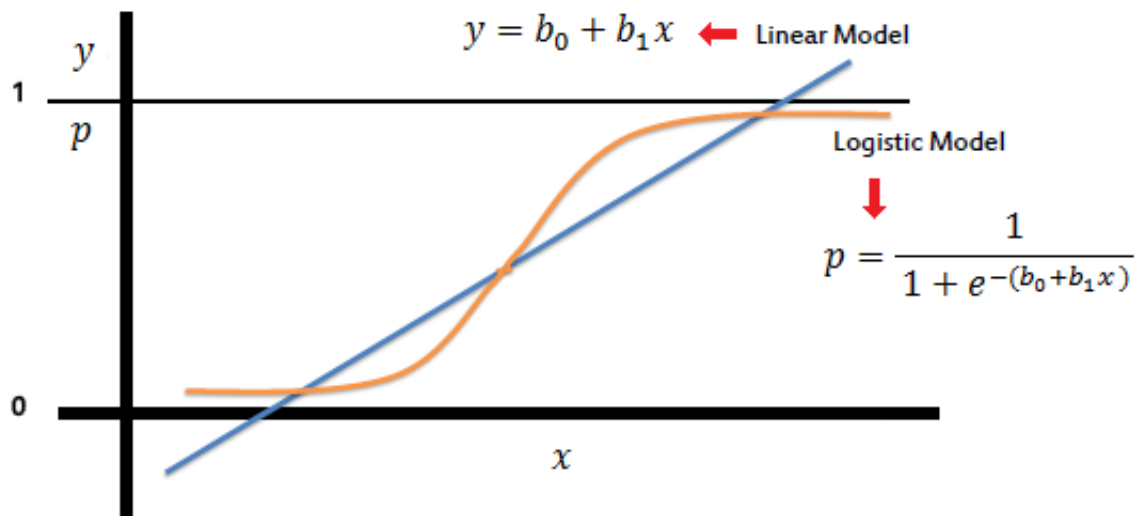
- Linear regression model can generate the **predicted probability**.
- As any number ranging from **negative to positive infinity**, whereas probability of an outcome can only lie between $0 < P(x) < 1$.



Also, Linear regression has a considerable effect on outliers.

- To avoid this problem, **log-odds function** or **logit function** is used.
- The logit is **not normally distributed** and **the variance is not constant**.

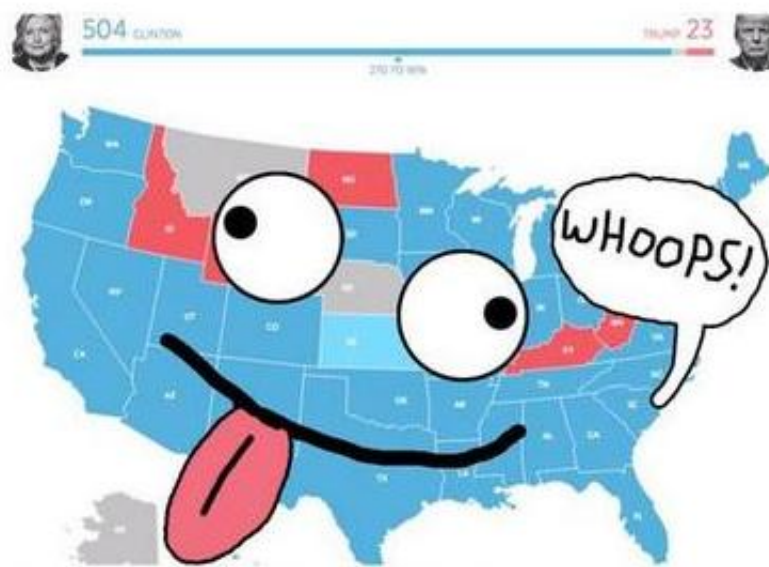
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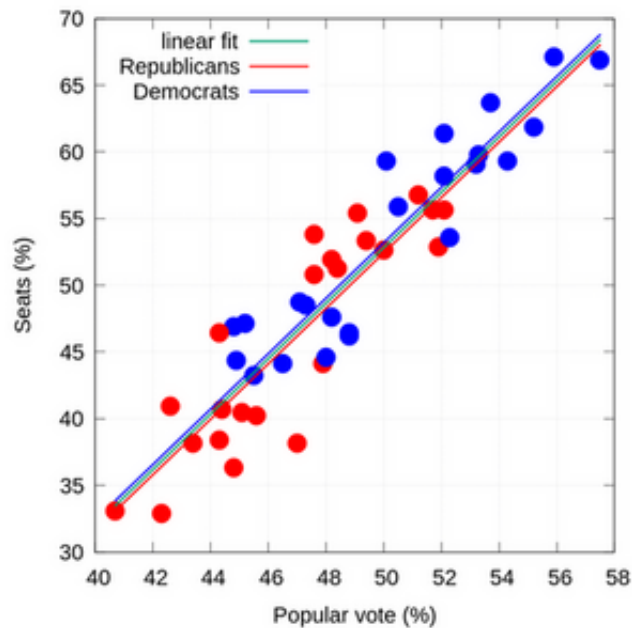
- Therefore, logistic regression requires a **more computationally** complex estimation method named as Method of Maximum Likelihood (ML) to estimate the parameters.
- ML obtains the model coefficients that **relate predictors** to the **target**.

For example

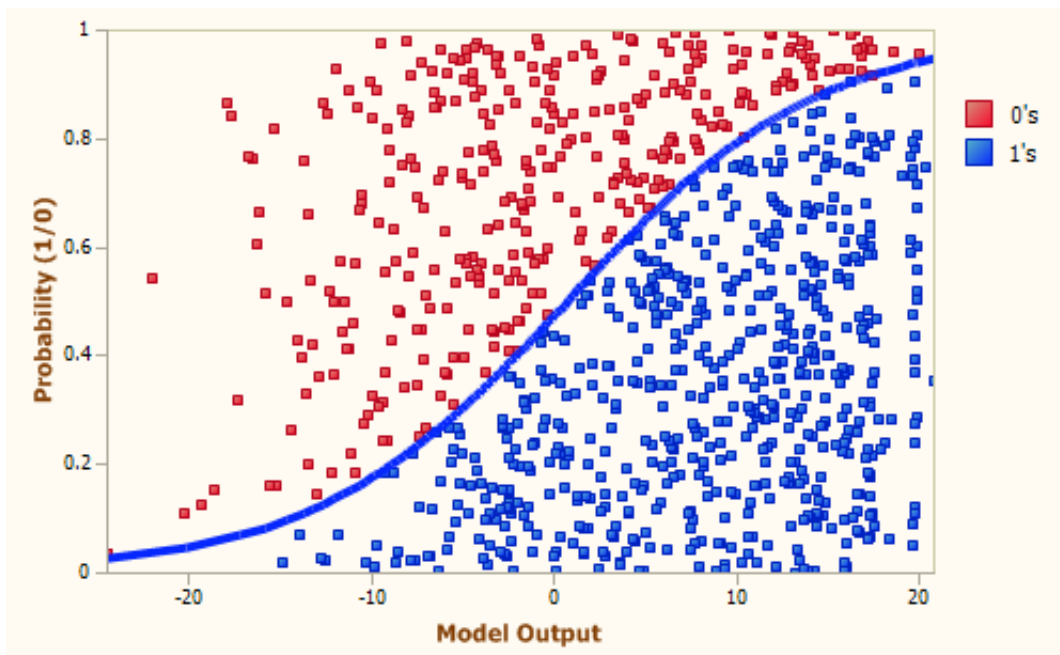
- Lets take a scenario where you are analyzing the **voting patterns of USA** to predict who will win the next election.



- **Linear Regression** - if you want to predict the **number of people**(continuous response) who will vote for **democrats/republicans** in each county/city/state etc.



- **Logistic Regression** - if you want to predict the probability that a certain **person** will vote for a **democrat/republican** or not.



What is Logit Function?

- Logistic regression can be expressed as:

The left hand side is called the **logit or log-odds function**, and $p(x)/(1-p(x))$ is called odds.

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$$

$$p = \frac{1}{1 + e^{-\text{logit}(p)}}$$

P is the probability of the event

ln is the natural log (to the base e)

Logit is also denoted as Ln

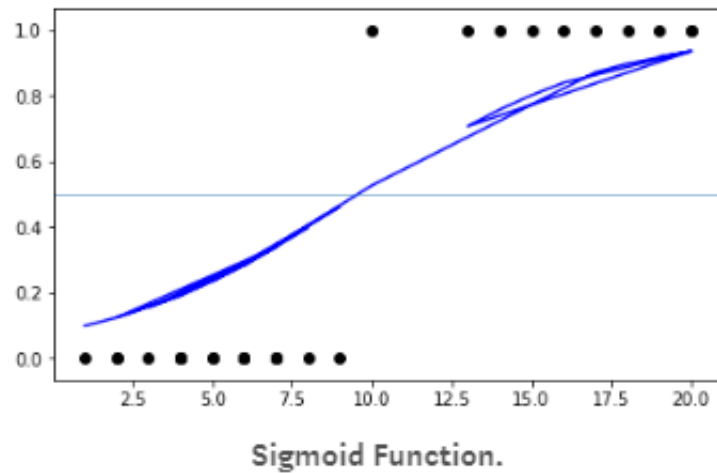
Logistic Regression formula is :

$$\text{Ln}\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

- The *odds* signifies the **ratio of probability** of success to **probability of failure**.
- Therefore, in Logistic Regression, linear combination of inputs are mapped to the **log(odds)** - the output being **equal to 1**.
- If we take an **inverse of the above function**, we get:

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}.$$

- This is known as the **Sigmoid function** and it gives an S-shaped curve.
- It always gives a **value of probability** ranging from **0 < p < 1**.



Multi-class Logistic Regression

- The basic intuition behind **Multi-class** and **binary Logistic regression** is same.
- However, **for multi-class problem we follow a one vs all approach.**
- For Example:
 - If we have to predict whether the weather is **sunny, rainy, or windy**, we are dealing with a Multi-class problem.



- We turn this problem into **three binary classification problem** i.e whether it is **sunny** or not, whether it is **rainy** or not and whether it is **windy** or not.
- We run all three classifications **independently** on input.

- The classification for which the **value of probability** is **maximum** relative to others, is the solution.

Advantages of Logistic Regression

- **Simple and efficient.**
- **Low variance.**
- It provides probability score for observations.

Disadvantages of Logistic Regression

- Doesn't handle large number of **categorical features/variables** well.
- It requires transformation of **non-linear features**.

Real Life Examples

1. Suppose the shop owner would like to **predict the customer** who entered into the shop will buy the **Macbook or Not**.
 - To predict whether the customer will **buy the MacBook or not**.
 - The shop owner will observe the **customer features** like.
- **Gender:**
 - Probabilities wise male will **high chances** of purchasing a MacBook than females.
- **Age:**
 - Kids **won't purchase** MacBook.
- The shop owner will use the above, **similar kind of features** to predict the **likelihood occurrence** of the event (Will buy the Macbook or not.)

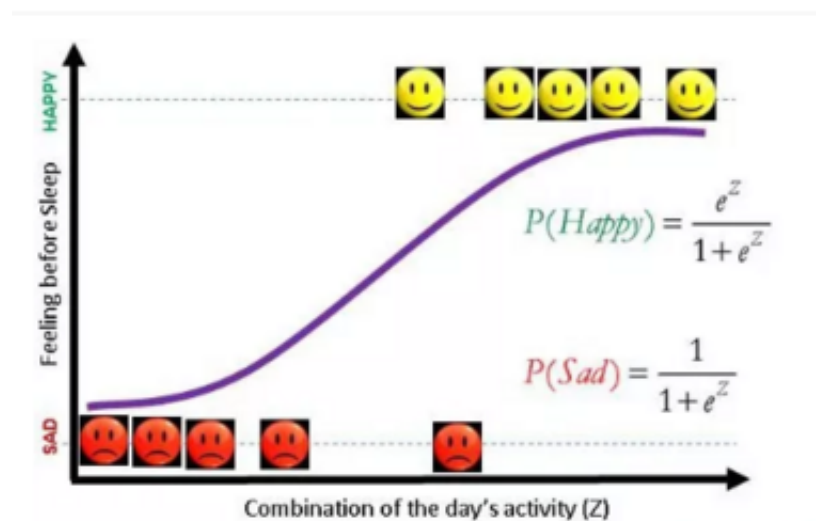


2. Suppose **Penguin** wants to know, how likely it will be happy based on its daily activities.
- If the penguin wants to build a logistic regression model to predict it **happiness based on its daily activities**.
 - The penguin needs both the **happy** and **sad activities**.
 - In machine Learning terminology these activities are known as the Input parameters **features** .



So let's create a table which contains penguin activities and the result of that activity like **happy** or **sad**.

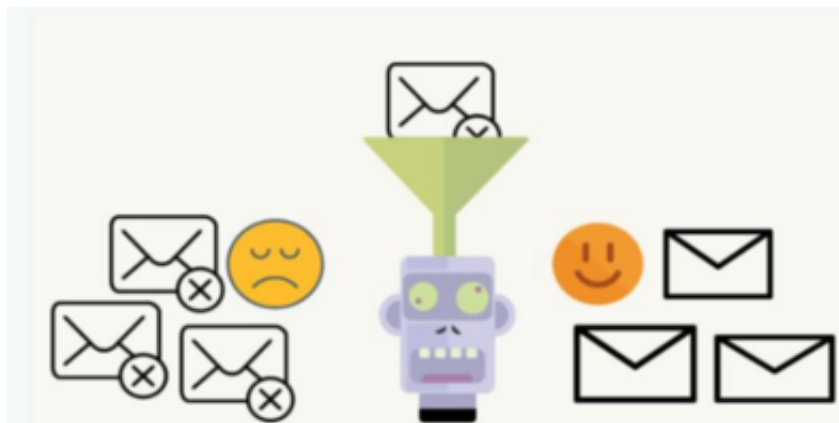
No.	Penguin Activity	Penguin Activity Description	How Penguin felt (Target)
1	X1	Eating squids	Happy
2	X2	Eating small Fishes	Happy
3	X3	Hit by other Penguin	Sad
4	X4	Eating Crabs	Sad



- Penguin is going to use the above **activities** (features) to train the logistic regression model.
- Later the trained logistic regression model will **predict** how the **penguin is feeling** for the new penguin activities.

3. Suppose you want to know whether the **email is a spam** or **not** Understand step by step working of the spam filter.

- Also, next time when you see a **“You have won a lottery”** email rather than ignoring it, you might prefer to report it as a spam.



- This **process** can help a lot of other people who are receiving the **same kind of email** but not aware of what spam is.
- Sometimes **wrong spam tag** can move a **genuine email** to **spam folder** too.
- So, you have to be careful before **you tag an email as a spam** or **not spam**.